

AP Calc Lesson 2.2 Homework

Answer
Key

1. $y - 5 = \frac{1}{3}(x + 2)$ or equivalent

2. a.
$$\begin{aligned}\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} &= \lim_{h \rightarrow 0} \frac{(5(x+h)^2 - 2(x+h)) - (5x^2 - 2x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{5x^2 + 10xh + 5h^2 - 2x - 2h - 5x^2 + 2x}{h} = \lim_{h \rightarrow 0} \frac{10xh + 5h^2 - 2h}{h} \\ &= \lim_{h \rightarrow 0} \frac{h(10x + 5h - 2)}{h} = \lim_{h \rightarrow 0} (10x + 5h - 2) = 10x - 2\end{aligned}$$

$$f'(x) = 10x - 2$$

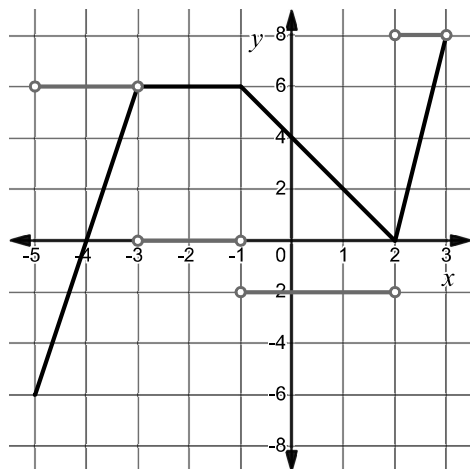
b. $f'(6) = 58$

c. The line tangent to the graph of f at $x = 6$ has a slope of 58.

3. a. $f(4) = 18$. Since the point of tangency is shared by both the tangent line and the curve, $f(4) = 5(4) - 2 = 18$.

b. $f'(4) = 5$. The tangent line at $x = 4$ has the same slope as the curve at $x = 4$, so $f'(4) = 5$.

4.



5. a. $f'(5) = 3$

b. $f'(x) = \frac{2}{5}x + 1$

c. $y - 30 = 5(x - 10)$

d. $f(x) = \frac{1}{5}x^2 + x$

6. a. At $x = 1$, the slope of the curve f is 2.
b. Answers will vary. Values greater than 5 but less than 5.5 are acceptable.
c. The graph of f is steeper at $x = 9$ because the slope of f is 6, whereas the slope of f at $x = 1$ is only 2.
d. $f'(4) = 4$, so the slope of the line normal to the graph of f at $x = 4$ is $-\frac{1}{4}$.

7. a. $f(0) = 2$
b. $f'(0) = -\frac{3}{5}$
c. $f(5) = -1$

8. $b(x + 2) = 2b(x + 1) + 1$ when $x = -\frac{1}{2} \rightarrow \frac{3}{2}b = b + 1 \rightarrow b = 2$

$$f'(-1) = b = 2 \text{ and } f'(0) = 2b = 4$$

9. C

10. B